# Kenechukwu Nwodo

nwodok@vt.edu

Address: Arlington, VA.

Website | LinkedIn | Google Scholar


**Summary:** Computer Engineering Ph.D. Candidate with research experience in agentic AI for network defense and LLM-driven vulnerability management, with industry experience in mobile security.

## Education

- **Virginia Tech** — Alexandria, VA
  *Ph.D. in Computer Engineering; GPA: 3.80* — *08/2020 - (Exp.)12/2026*
  - **Advisors**: Dr. Angelos Stavrou and Dr. Haining Wang
  - **Dissertation Title**: Securing Software Systems and Autonomous Network Defense
- **Virginia Tech** — Arlington, VA
  *M.S. in Computer Engineering; GPA: 3.78* — *08/2020 - 05/2023*
- **Rensselaer Polytechnic Institute** — Troy, NY
  *B.S. in Electrical Engineering* — *08/2016 - 05/2020*

## Research Experience

- **Network and Systems Security Lab, Virginia Tech** — Alexandria, VA
  *Graduate Research Assistant* — *08/2020 - Present*
  - **Autonomous Cyber Agent Training and Evaluation**: Evaluating heuristic and RL-based defensive and attacker agents across simulation and emulation environments to generate insights for remediation and develop stronger, more adaptive defensive agents.
  - **Software Vulnerability Management**: Reconstructing CVE report descriptions with multiple LLM-developed configurations to facilitate easier automation of vulnerability analysis, mapping vulnerabilities from CVEs to MITRE ATT&CK techniques.
  - **Vulnerability Detection**: Automating the identification of vulnerable libraries and packages from Software Bill of Materials in 5G virtual network functions.
  - **SCADA System/IoT Security**: Detecting and mitigating wireless attacks (DoS, Code Injection) on IoT devices and SCADA systems; with the BEMOSS open-source platform, using a shadow replica.
  - **Embedded Systems Security**: Developing a mutual authentication system with STSAFE for FIDO-like applications.
  - **Sensing for Smart Home Monitoring**: Partnering with the School of Construction to create software and hardware for environmental sensor data collection on Raspberry Pis for deployment in Virginia homes.

## Industrial Experience

- **Quokka (Formerly Kryptowire Inc)** — McLean, VA
  *Research Engineer Intern* — *06/2022 - 12/2022*
  - Designed a novel static analysis tool to automatically generate a software bill of materials of libraries (names + versions) that comprise an Android app when obfuscation is applied, and performed a comparative analysis.
- **Boeing** — Seattle, WA
  *Hardware-Electronics Electrical Engineering Intern* — *05/2019 - 08/2019*
  - Created software to give 3D positional system orientation and height measurements using multiple LIDARs for Boeing planes. Also investigated 4G-LTE as a high bit-rate data communication link.
- **SelfArray** — Troy, NY
  *Computer/Electrical Engineering Intern* — *06/2018 - 08/2018*
  - Designed pattern-detection software using Python and OpenCV to test the accuracy and efficiency of the company's patented self-assembly method.

## Publications

- **Kenechukwu Nwodo**, et al. [Full Title Withheld, Under Blind Review] "Cyber Agent Training & Evaluation." *Under Review at Conference, 2026.*
- **Kenechukwu Nwodo**, et al. [Full Title Withheld, Under Blind Review] "Cyber Agent Evaluation Study." *Under Review at Conference, 2026.*
- **Kenechukwu Nwodo**, Angelos Stavrou. "Software Engineering Reliability in the Era of Generic and Agentic AI" *Chapter accepted for publication in Generative and Agentic AI Reliability: Architectures, Challenges, and Trust for Autonomous Systems, Springer Nature, August 5, 2026.* (Link to Springer page)
- **Kenechukwu Nwodo**, Sudip Maitra, Shihua Sun, Haining Wang, Angelos Stavrou. "Improving Security Vulnerability Descriptions using LLMs" *Chapter accepted for publication in Generative and Agentic AI Reliability: Architectures, Challenges, and Trust for Autonomous Systems, Springer Nature, August 5, 2026.* (Link to Springer page)
- Sudip Maitra, **Kenechukwu Nwodo**, Tolga Atalay, Angelos Stavrou, Haining Wang. "Towards Securing Access Control in 5G and Beyond with Zero Trust" *Paper accepted for publication in ACM Conference on Data and Application Security and Privacy (CODASPY), 2026.*
- Shihua Sun, Pragya Sharma, **Kenechukwu Nwodo**, Angelos Stavrou, Haining Wang. "FedMADE: Robust Federated Learning for Intrusion Detection in IoT Networks Using a Dynamic Aggregation Method" *International Conference on Information Security (ISC), 2024.*
- Shihua Sun, **Kenechukwu Nwodo**, Shridatt Sugrim, Angelos Stavrou, Haining Wang. "ViTGuard: Attention-aware Detection against Adversarial Examples for Vision Transformer" *Annual Computer Security Applications Conference (ACSAC), 2024.*
- David A Tomassi, **Kenechukwu Nwodo**, Mohamed Elsabagh. "Libra: Library Identification in Obfuscated Android Apps" *International Conference on Information Security (ISC), 2023.*
- **Kenechukwu Nwodo**, et al. "Towards Accurate and Reliable Industrial Intrusion Detection Systems Using Shadow Replicas" *MS Thesis, 2023.*

## Talks and Presentations

- **TALK: Improving Security Vulnerability Descriptions using LLMs**
  Presented at DEF CON 34 - Blacks In Cybersecurity Village at Las Vegas, NV, *08/08/2026* (Link to talk schedule)
- **POSTER: Bridging Simulation and Emulation for Autonomous Cyber Agent Training and Evaluation**
  Presented at the 35th USENIX Security Symposium 2026 at Baltimore, MD, *08/12/2026* (Link to poster schedule)

## Professional Service

- **Reviewer**: IEEE International Conference on Computing and Machine Intelligence, *2024 & 2025*
- **Reviewer**: Commonwealth Cyber Initiative's Central Virginia Node FY27 Proposals for Research Grant & Workforce Development Grant, *2026*
- **Volunteer**: DEF CON 34 - Blacks in Cybersecurity Village, Las Vegas, NV, *2026*
- **Volunteer Organizer**: IEEE Reliability Society "AI Trusted Data" Workshop, Alexandria, VA, *2026*

## Teaching Experience

- **Intro to Computer Networking - ECE 3564, Virginia Tech**: Graduate Teaching Assistant, *01/2022 - 05/2022*
  Held weekly office hours, proctored exams, supervised two graders, and delivered a lecture on Public Key Cryptography.

## Course Projects

- **Advanced Machine Learning, Fall 2021**: Implemented NB and SVM supervised machine learning classifiers to perform Intrusion Detection for Industrial Control System networks.
- **Network Security, Spring 2021**: Demonstrated the key reinstallation attack (WPA2 attack) on vulnerable and protected Android phones.
- **Network Architecture and Protocols, Fall 2020**: Performed a comparative analysis on 4G-LTE and 5G, for an islanded microgrid network.
- **CAPSTONE, Fall 2019**: Developed a noise dosimeter with email notifications for a server utilizing Django.

## Skills & Awards

- **Research Skills**: Reinforcement Learning, Simulation, Emulation, MITRE ATT&CK, Prompt Engineering, Vulnerability Detection, Static Analysis, Android Library Identification, Network Hardening, SCADA/IoT Security, Traffic Analysis, ML Applications
- **Relevant Courses**: Network Security, Fundamentals of InfoSec, Network Architecture and Protocols, Advanced Machine Learning, System and Software Security, Optimization Techniques, Object Oriented Programming
- **Programming Languages, Tools, and Operating Systems**: Python, Java, C, Linux (Fedora, Ubuntu, Kali, Raspbian), GPU Servers, vLLM, Ollama, LangChain, Prompt Engineering, PyTorch, TensorFlow, HuggingFace, Git, Docker, Apache Cassandra
- **Affiliations**: IEEE Member, ACM Member, USENIX Member, CEISCA (*2024 - Present*), iMentor (CCS *2020*)
- **Grants**: USENIX Security Grants for Black Computer Science Students (*2023*), GREPSEC VI (*2023*)
- **Awards**: Davenport Fellowship by ECE, Virginia Tech (*Spring 2026*)